

# Average (Homework)

Find the arithmetic mean of each data set below.

① 10.3, 0.8, 1.2, 5.6

② 9, 3, 5, 7, 10, 6

Find the geometric mean of each data set below.  
Round to the nearest ten thousandth.

③ 1, 2, 1, 2.4, 1

④  $\frac{2}{5}, \frac{3}{4}, \frac{7}{8}$

Find the median of each data set below.

⑤ 100, 80, 79, 86, 90, 81, 98, 92

⑥ -10, -5, -6, -5, -9, -7, -9

Find the mode of each data set below.

⑦ 1, 1, 2, 3, 1, 3, 3, 3, 2, 3

⑧ 5, 6, 10, 6, 4, 10, 5, 5

⑨ Bill sold hats, t-shirts, and buttons at the football stadium yesterday. Use the table below to calculate the average price per item of all the things that he sold.  
Round to the nearest cent.

| Item    | # sold | Price/item(\$) |
|---------|--------|----------------|
| Hat     | 21     | 8              |
| t-shirt | 100    | 20             |
| button  | 35     | 2              |

⑩ The table below shows information about all the car wheels sold by a company last year. What is the average diameter of the wheels sold by this company?  
Round to the nearest hundredth.

| Brand  | # sold | Diameter(in) |
|--------|--------|--------------|
| Rocko  | 10     | 20           |
| Jenson | 10     | 15           |
| Altar  | 30     | 18           |
| Lobo   | 28     | 17           |



⑪ 69.09% of the copper atoms on earth have a mass of 62.93 amu. 30.91% of the copper atoms have a mass of 64.9278 amu. What is the average mass of the copper atoms on earth? Round to the nearest hundredth.

⑫ 75.53% of the chlorine atoms on earth have a mass of 34.968 amu. 24.47% of the chlorine atoms have a mass of 36.956 amu. What is the average mass of the chlorine atoms on earth? Round to the nearest hundredth.

⑬ Billy walked 380 yards during his 35 minute hike. What was his average speed during the whole hike?

During the hike, Billy stopped to tie his shoes and to use the drinking fountain. If the total time that he walked was 32 minutes, what was his average speed during the time that he was walking?

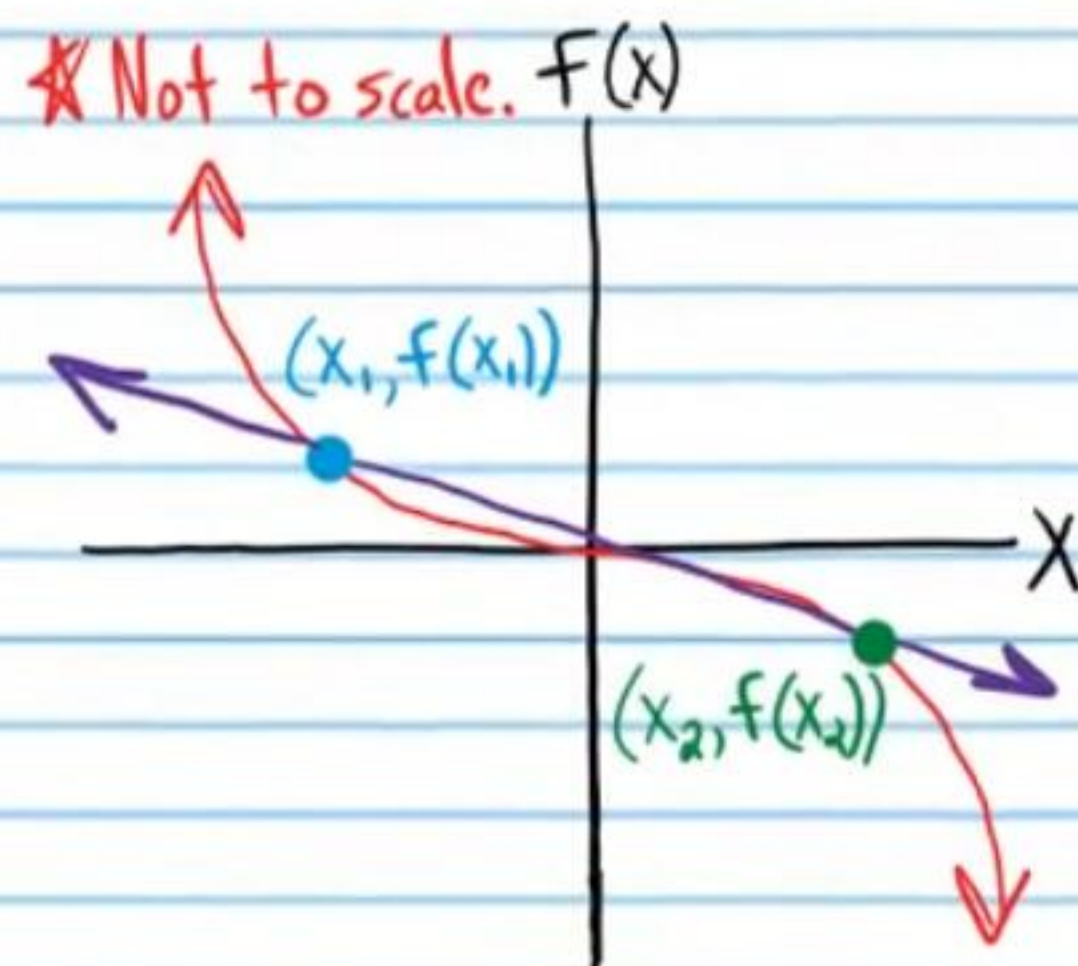
⑭ Ronald got a job washing dishes. He washed 42 plates on his first day at work. He washed 20 plates on his second day of work. If he washed dishes for 4 hours during his first day at work and 1 hour during his second day at work, what is the average number of plates that he washed per hour during the time that he washed dishes?

Ronald did other things on his first two days at work (e.g. sweeping the floor). If he worked a full eight hour shift each day, what is the average number of plates that he washed per hour during the 16 hours that he was on the job?

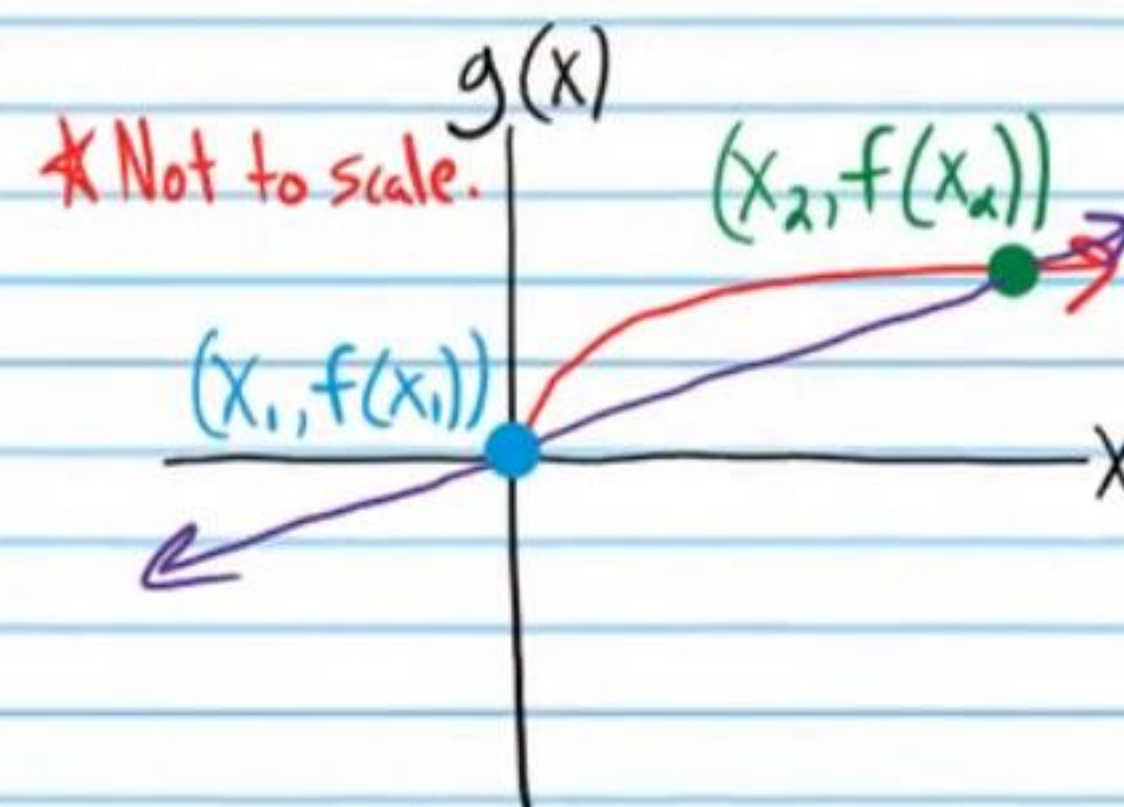
⑮ Tommy drove his car 82 Kilometers per hour for 5 hours. He then turned around and drove 60 Kilometers per hour for 1.5 hours. What was his average speed? Round to the nearest thousandth.



⑩ If  $f(x) = -\frac{1}{4}x^3$ , what is the average rate of change of the function per unit  $x$  from  $x = -1$  to  $x = 1$ ?



⑪ If  $g(x) = \sqrt{x}$ , what is the average rate of change of the function per unit  $x$  from  $x = 0$  to  $x = 4$ ?





Given the following functions, find algebraic expressions for the values below.

⑱  $f(x) = x^2$

i)  $f(c-1)$

ii)  $f(-\frac{5a}{3})$

iii)  $f(10x) - f(9x)$

⑲  $g(x) = \frac{1}{\sqrt{x}-x}$

i)  $g(d)$

ii)  $g(x^3)$

iii)  $g(n) - g(t)$

⑳  $h(x) = \frac{x-1}{x+1}$

i)  $h(x+1)$

ii)  $h(x-1) + h(x+1)$

For each function, find an algebraic expression representing the difference quotient (use "h" for change in "x"). It is not necessary to simplify each expression.

㉑  $f(x) = 2x+1$

㉒  $f(x) = |x^3|$

㉓  $f(x) = x \log_9 x$



$$(24) f(x) = \cot(x-4) + 3x$$

$$(25) f(x) = \frac{1}{x^2 + 2x - 4}$$

$$(26) f(x) = \frac{x}{2} + \sec x^2 - |x|$$

$$(27) f(x) = -\frac{1}{2}x - x^5$$

$$(28) f(x) = \frac{2}{x} - \csc x$$



# Homework Answers

① 4.475

⑭  $12\frac{2}{5}$  Plates/h or 12.4 Plates/h

②  $6\frac{2}{3}$

$3\frac{7}{8}$  Plates/h or 3.875 plates/h

③ 1.3685

⑮ 76.923 km/h

④ 0.6403

⑯  $-\frac{1}{4}$

⑤ 88

⑥ -7

⑦ 3

⑧ 5

⑨ \$14.35/item

⑩ 17.51 in

⑪ 63.55 amu

⑫ 35.45 amu

⑬  $10\frac{6}{7}$  yds/min or 10.857 yds/min

$11\frac{7}{8}$  yds/min or 11.875 yds/min

ii)  $(-\frac{5a}{3})^2 = \frac{25a^2}{9}$

iii)  $(10x)^2 - (9x)^2 = 100x^2 - 81x^2 = 19x^2$

⑲ i)  $\frac{1}{\sqrt{d}-d}$

⑳  $\frac{\frac{1}{(x+h)^2+2(x+h)-4} - \frac{1}{x^2+2x-4}}{h}$

ii)  $\frac{1}{\sqrt{x^3}-x^3}$

iii)  $\frac{1}{\sqrt{n}-n} - \frac{1}{\sqrt{t}-t}$

㉑  $\frac{\frac{x+h}{2} + \sec(x+h)^2 - |x+h| - (\frac{x}{2} + \sec x^2 - |x|)}{h}$

㉒ i)  $\frac{x}{x+2}$

㉓  $\frac{-\frac{1}{2}(x+h) - (x+h)^5 - (-\frac{1}{2}x - x^5)}{h}$

ii)  $\frac{x-2}{x} + \frac{x}{x+2}$

㉔  $\frac{\frac{2}{x+h} - \csc(x+h) - (\frac{2}{x} - \csc x)}{h}$

㉕  $\frac{2(x+h)+1 - (2x+1)}{h}$

㉖  $\frac{|(x+h)^3| - |x^3|}{h}$

㉗  $\frac{(x+h)\log_9(x+h) - x\log_9 x}{h}$

㉘  $\frac{\cot(x+h-4) + 3(x+h) - (\cot(x-4) + 3x)}{h}$